DAWOOD AHMAD

Remote Sensing Data Engineer | Machine Learning Engineer | GeoAI

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PROFILE

Machine Learning Engineer and Team Lead who builds **operational satellite data pipelines**, from raw imagery to production map products. I lead a team that runs crop-monitoring systems on **Sentinel-1 SAR, Sentinel-2 and Landsat** time series for agribusinesses and development agencies in Asia, Africa and South America. My daily work is Python geospatial code, PostgreSQL/PostGIS, Git, Docker, model training and data quality checks that catch errors before they reach the client. I also co-authored a study of **carbon monoxide from Sentinel-5P TROPOMI** over Pakistan, so I have worked with atmospheric trace-gas data as well. M.Phil. in Space Science (Gold Medalist), two peer-reviewed papers.

TECHNICAL SKILLS

Programming and data: Python (NumPy, Pandas, GDAL, Rasterio, GeoPandas, Shapely), SQL, PostgreSQL/PostGIS, Docker, R, Git/GitHub, Linux.

Remote sensing: Sentinel-1 SAR, Sentinel-2, Landsat, Sentinel-5P TROPOMI, MODIS. Google Earth Engine, STAC, QGIS, ArcGIS Pro, ERDAS IMAGINE.

Machine learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost. Time-series classification, semantic segmentation, Segment Anything (SAMGeo), super-resolution.

Data quality and validation: grouped spatial and temporal validation, leakage detection, domain-shift checks, accuracy assessment against field surveys.

Web and services: Django, GeoServer, Leaflet, OpenLayers, ODK Central, real-time GIS dashboards.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer and Team Lead — **March 2025 to Present**

Farmdar, Lahore, Pakistan — *Agri-tech and Earth observation*

- **Team leadership:** lead a team of machine-learning and remote-sensing engineers who deliver satellite crop-monitoring products to clients in several countries.
- **National-scale SAR processing (Thailand):** building a country-wide sugarcane map of Thailand from Sentinel-1 SAR time series, with radar data processed at national scale.
- **End-to-end production pipelines:** built the pipeline around the SAMGeo foundation model for field-boundary delineation, from imagery preparation to cleaned vector polygons, now used for sugarcane mapping in Thailand and South Africa.
- **Multi-year map production:** built the training data, models and production pipeline for district-level crop-type maps of Punjab and Sindh, Pakistan, for every year from 2014 to 2025, harmonising Sentinel-2 and Landsat 8 time series.
- **Multi-sensor method design (Myanmar):** designed a label-free rice-mapping method where Sentinel-2 crop-calendar rules find rice and the Sentinel-1 flooding signal confirms it, plus a radar-only model for cloudy seasons. Maps agreed closely with independent field surveys.
- **Data quality and debugging:** found that a random pixel split was hiding a sugarcane model's failure on new acquisition dates and rebuilt it with grouped validation and a domain-shift guard. Traced an inflated cotton area estimate to a time-window bug in the pipeline.
- **SAR research (Philippines):** lead the Sentinel-1 research on maize detection in a persistently cloudy tropical climate.
- **Time-series products (Pakistan, Brazil):** multi-year NDVI models for sugarcane, with an NDVI-trough method that detects sowing and harvest windows.

GIS Developer — **December 2024 to March 2025**

NESPAK, Lahore, Pakistan — *Public-sector engineering consultancy*

- Built a real-time GIS progress-monitoring dashboard for the WASA Lahore Development Project, with field-survey data flowing in through ODK Central.

Data Processing Analyst **July 2024 to November 2024**

Geospatial Planning and Engineering Associates, Lahore, Pakistan *Geospatial services*

- Built high-resolution image classifiers and a deep-learning model for satellite-image super-resolution. Managed imagery quality assessment and ground-control planning.

Geospatial Intern **May 2024 to July 2024**

Farmdar, Lahore, Pakistan *Agri-tech and Earth observation*

- Wrote Sentinel-2 and Google Earth Engine crop-classification scripts and an initial maize-detection model, which led to a full-time role.

EDUCATION

M.Phil. in Space Science **2024 to 2026**

Institute of Space Science, University of the Punjab, Lahore *Gold Medalist, CGPA 3.98/4.00*

- Degree completed in 2026. Thesis: comparative evaluation of Sentinel-1 SAR and Sentinel-2 optical data for crop classification, including computational trade-offs.

B.S. in Space Science **2020 to 2024**

Department of Space Science, University of the Punjab, Lahore *CGPA 3.83/4.00*

- Final-year project: end-to-end WebGIS application with Leaflet, GeoServer, PostgreSQL/PostGIS and Django.

PUBLICATIONS

- **Ahmad, D.**, & Haseeb, M. (2026). Machine Learning and Deep Learning for Snowmelt Prediction: Long-Term Trends and Implications for Water Resource Management. *Water Resources Management*, 40(9), 407. DOI: 10.1007/s11269-026-04775-0
- Xue, L., Khan, A., Haseeb, M., Aqnouy, M., **Ahmad, D.**, Ghodhbani, R., et al. (2025). Multi-Sensor Fusion and Deep Learning Approaches for Semantic Segmentation of Glacial Lakes. *IEEE JSTARS*. DOI: 10.1109/JSTARS.2025.3576187
- Bano, S., Javed, A. M., Mahmood, K., **Ahmad, D.**, Tariq, S., & Parvez, S. GeoAI-based Spatio-Temporal Dynamics and Explainable AI Modelling of Carbon Monoxide over Pakistan using Sentinel-5P TROPOMI (2018 to 2026). *Under review.*
- Two further manuscripts under review on Sentinel-1/Sentinel-2 rice classification and operational maize mapping in Punjab, Pakistan.

LANGUAGES

English: professional working proficiency. **Urdu:** native. **Punjabi:** native.

REFERENCES

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